Artificial Intelligence and Neural Networks

What is Artificial Intelligence?

Artificial Intelligence is the simulation of human intelligence processes by machines, especially computer systems. It includes learning, reasoning, and self-correction.

Neural Networks Explained

Neural networks are computing systems inspired by biological neural networks in the human brain. They consist of layers of interconnected nodes called neurons.

An input layer receives raw data, hidden layers process the information, and the output layer produces the final result or prediction.

Deep Learning

Deep Learning is a subset of machine learning that uses neural networks with many hidden layers to learn complex representations of data.

Convolutional Neural Networks (CNNs) are used for image recognition. Recurrent Neural Networks (RNNs) are used for sequential data like text and speech.

Activation Functions

Activation functions determine whether a neuron should be activated. Common functions include ReLU, Sigmoid, and Tanh.

The ReLU function outputs zero for negative values and the input value for positive values, making it computationally efficient.

Training Neural Networks

Neural networks are trained using backpropagation and gradient descent to minimize the loss function and improve prediction accuracy over time.